

The One-Dimensional-Kernel Case of the Minor-Invariant Projection Question

Run fa_banach_001, lane 19

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Source Question

Guerra and Raita ask in Question 5.17 of arXiv:1909.03923 for conditions on a projection

$$T : \mathbb{R}^{m \times n} \rightarrow V \subseteq \mathbb{R}^{m \times n}$$

under which one can find a nonzero linear combination H_s of s -minors, $2 \leq s < \min\{m, n\}$, satisfying

$$H_s(X) = H_s(X + Y) \quad \text{for every } X \text{ and every } Y \text{ with } T(Y) = 0.$$

Equivalently, H_s is a linear combination of s -minors depending only on the coordinates of V . The source passage is shown in Figure 1.

This packet gives an exact answer when $\ker T$ is one-dimensional.

Definitions and Normal Form

For $I \subset \{1, \dots, m\}$ and $J \subset \{1, \dots, n\}$ with $|I| = |J| = s$, write

$$M_{I,J}(X) = \det X_{I,J},$$

where $X_{I,J}$ is the submatrix with rows I and columns J , both listed in increasing order. A linear combination of s -minors is a polynomial

$$H(X) = \sum_{\substack{|I|=s \\ |J|=s}} c_{I,J} M_{I,J}(X).$$

If $\ker T = \mathbb{R}A$, then the source condition is exactly

$$H(X + tA) = H(X) \quad \text{for all } X \in \mathbb{R}^{m \times n}, \quad t \in \mathbb{R}.$$

Let $r = \text{rank } A$. By invertible row and column operations there are $P \in GL_m(\mathbb{R})$ and $Q \in GL_n(\mathbb{R})$ such that

$$PAQ = D_r,$$

where D_r is the $m \times n$ matrix with ones in the entries $(1, 1), \dots, (r, r)$ and zeros elsewhere. By the Cauchy-Binet formula, $X \mapsto K(PXQ)$ and $X \mapsto K(P^{-1}XQ^{-1})$ carry the vector space spanned by s -minors onto itself. Thus the existence of a nonzero invariant combination for the direction A is equivalent to the same existence problem for the normal-form direction D_r .

div-null Lagrangians.

It would be interesting to give a theoretical characterization of the solutions of the computational problem. This is also a relevant question since the linear system described above grows factorially in $\dim \mathbb{V}$, although in applications to continuum mechanics this number is usually relatively small. Unfortunately, even in the special case when $\mathcal{B}$ has order one such a characterization seems difficult. The authors were unable to give a definitive answer even to the following simple-looking question.

Assume we are given a projection $T: \mathbb{R}^{m \times n} \rightarrow \mathbb{V}$, which can be chosen to be orthogonal, onto some subspace $\mathbb{V} \subseteq \mathbb{R}^{m \times n}$. Consider a function $H_s: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ as above, i.e.

$$H_s(X) = \sum_{\deg M=s} c_M M(X), \quad H_s = H_s \circ T$$

where the sum runs over the set of all minors (not necessarily principal) of order $2 \leq s < \min\{m, n\}$. The second condition can be equivalently rewritten as

$$(5.16) \quad H_s(X) = H_s(X + Y) \text{ for all } X, \text{ all } Y \text{ such that } T(Y) = 0.$$

We think of this as saying that the linear combination of minors H_s only depends on the coordinates of $\mathbb{V}$.

5.17 Question. Under which conditions on T can we find non-zero H_s satisfying (5.16)? Can we characterize such H_s in terms of $\mathbb{V}$?

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Figure 1: Question 5.17 in the source paper, printed page 24.

Proof Intuition

After the reduction to D_r , the kernel direction changes only the first r diagonal entries. Call these indices active. A single minor $M_{I,J}$ is automatically invariant if no active index belongs to both I and J , because then the shifted diagonal entries never occur inside the submatrix. Such row/column sets exist exactly when two s -sets can be placed in the m row slots and n column slots without double-using any of the r active indices; this is the counting condition $2s \leq m + n - r$.

In the opposite range, every s -row set and s -column set must share at least one active index. Differentiating the identity $H(X + tD_r) = H(X)$ gives one linear equation for each $(s-1)$ -minor. The equation for $M_{I \setminus \{a\}, J \setminus \{a\}}$ contains the coefficient of $M_{I,J}$. If one descends on the number of active indices appearing in $I \cup J$, the other coefficients in that same equation have already vanished, so $c_{I,J}$ is isolated and must be zero.

Main Partial Theorem

Theorem 1 (One-dimensional-kernel rank threshold). *Let $m, n \geq 2$, let $2 \leq s < \min\{m, n\}$, and let $T: \mathbb{R}^{m \times n} \rightarrow V$ be a linear projection with one-dimensional kernel $\ker T = \mathbb{R}A$, where $A \neq 0$ has rank r . Then there is a nonzero linear combination H of s -minors satisfying*

$$H(X) = H(X + Y) \quad \text{whenever } T(Y) = 0$$

if and only if

$$2s \leq m + n - r.$$

Proof. By the normal-form discussion above, it is enough to prove the theorem for $A = D_r$. Write

$$R = \{1, \dots, r\}$$

for the active indices.

First assume $2s \leq m + n - r$. This inequality says that it is possible to choose subsets $I \subset \{1, \dots, m\}$ and $J \subset \{1, \dots, n\}$ with $|I| = |J| = s$ and

$$I \cap J \cap R = \emptyset.$$

Indeed, choose I with as few active indices as possible. Then

$$|I \cap R| = \max\{0, s - (m - r)\}.$$

If $s \leq m - r$, all active column labels remain available and there is no obstruction to choosing J . If $s > m - r$, the number of column labels not forbidden by $I \cap R$ is

$$n - |I \cap R| = n - s + m - r,$$

which is at least s precisely because $2s \leq m + n - r$. Thus such I, J exist. For such I, J , adding tD_r changes only entries (a, a) with $a \in R$, and none of these entries lies in the submatrix $X_{I,J}$. Hence $M_{I,J}(X + tD_r) = M_{I,J}(X)$ for all X, t , giving a nonzero invariant s -minor.

Conversely, assume $2s > m + n - r$. Then every pair of subsets $I \subset \{1, \dots, m\}$ and $J \subset \{1, \dots, n\}$ with $|I| = |J| = s$ satisfies

$$I \cap J \cap R \neq \emptyset.$$

Let

$$H(X) = \sum_{\substack{|I|=s \\ |J|=s}} c_{I,J} M_{I,J}(X)$$

be invariant under $X \mapsto X + tD_r$. Differentiating at $t = 0$ gives the polynomial identity

$$0 = \left. \frac{d}{dt} \right|_{t=0} H(X + tD_r).$$

For one minor,

$$\left. \frac{d}{dt} \right|_{t=0} M_{I,J}(X + tD_r) = \sum_{a \in I \cap J \cap R} \varepsilon(a; I, J) M_{I \setminus \{a\}, J \setminus \{a\}}(X),$$

where each $\varepsilon(a; I, J)$ is 1 or -1 .

The minors of a fixed order are linearly independent as polynomials: every monomial in $M_{I,J}$ uses exactly the row set I and the column set J . Therefore, for every pair I_0, J_0 of $(s-1)$ -subsets, the coefficient of M_{I_0, J_0} in the differentiated identity is zero:

$$\sum_{a \in R \setminus (I_0 \cup J_0)} \varepsilon(a; I_0 \cup \{a\}, J_0 \cup \{a\}) c_{I_0 \cup \{a\}, J_0 \cup \{a\}} = 0. \quad (1)$$

Here $I_0 \cup J_0$ is understood as a union of active labels when intersected with R ; equivalently the condition on a is $a \notin I_0$ and $a \notin J_0$.

We now prove that all coefficients vanish by descending induction on

$$N(I, J) = |(I \cup J) \cap R|.$$

The largest possible value is r . Suppose first that $N(I, J) = r$. Choose $a \in I \cap J \cap R$ and set $I_0 = I \setminus \{a\}$, $J_0 = J \setminus \{a\}$. Since $(I \cup J) \cap R = R$, we have

$$R \setminus (I_0 \cup J_0) = \{a\}.$$

Thus equation (1) contains only the term $\varepsilon(a; I, J)c_{I,J}$, so $c_{I,J} = 0$.

Now fix $N < r$ and assume all coefficients $c_{I',J'}$ with $N(I', J') > N$ have already been shown to vanish. Take I, J with $N(I, J) = N$. Choose $a \in I \cap J \cap R$ and again set $I_0 = I \setminus \{a\}$, $J_0 = J \setminus \{a\}$. In equation (1), the summand with added index a is precisely $\varepsilon(a; I, J)c_{I,J}$. Any other summand has added index $b \in R \setminus ((I \cup J) \cap R)$, so

$$N(I_0 \cup \{b\}, J_0 \cup \{b\}) = N + 1.$$

Its coefficient is zero by the induction hypothesis. Equation (1) therefore reduces to $\varepsilon(a; I, J)c_{I,J} = 0$, and hence $c_{I,J} = 0$.

Since every I, J in the range $2s > m+n-r$ has a common active index, the descending induction applies to every coefficient. Hence $H = 0$. This proves that no nonzero invariant combination exists in the opposite range. $\square$

Relation to the Earlier Trace-Free Packet

The previous packet

`1909.03923_tracefree_minor_invariant_threshold`

handled the square trace-free projection, where $\ker T = \mathbb{R}I_n$. In the notation of Theorem 1, this has $m = n$ and $r = n$, so the threshold becomes $2s \leq n$. Thus the present result strictly subsumes that slice while still remaining a partial answer to the full Question 5.17.

Scope and Novelty Check

The theorem gives an exact existence classification for all projections with one-dimensional kernel. It does not classify projections whose kernel has dimension at least two. In the positive range, it proves existence but does not compute the full invariant subspace.

Cheap run indexes were searched for arXiv:1909.03923, trace-free, scalar-shift, one-dimensional kernel, rank threshold, minor invariant, and the source question keywords. The only duplicate-adjacent hit was the earlier trace-free packet, which is subsumed here. A bounded web search on July 19, 2026 found the source paper and preprint records but no exact answer to this one-dimensional-kernel classification.

Computational Sanity Check

The script `code/check_onedim_kernel_threshold.py` constructs the exact rational matrix of the D_r -directional derivative on s -minors and checks the dimension of its kernel for all $3 \leq m, n \leq 6$, all $2 \leq s < \min\{m, n\}$, and all $1 \leq r \leq \min\{m, n\}$. It verifies that the kernel is nonzero exactly in the range $2s \leq m + n - r$. This computation is only a sanity check; the proof above is independent.

References

1. A. Guerra and B. Raita, *Quasiconvexity, null Lagrangians, and Hardy space integrability under constant rank constraints*, arXiv:1909.03923; Archive for Rational Mechanics and Analysis 245 (2022), 279–320.